Model-File

Large language models are general-purpose but in real life you want specialized behavior.

You might want:

- A polite customer support bot.
- A brutally honest code reviewer.
- A legal assistant that avoids giving medical advice.
- A casual Gen-Z style chatbot.

Brain $\xrightarrow{\text{LLM}}$ New model

existing model.

1. General purpose model → Learning
2. Instruction

A modelfile in Ollama is a configuration file that defines how to a model should behave, respond, and be used. It does not ~~onto~~ contain the AI model itself. Instead, it acts as a blueprint or instruction layer on top of an existing base model, where you take a base model and on that model give instructions.

A modelfile usually defines:

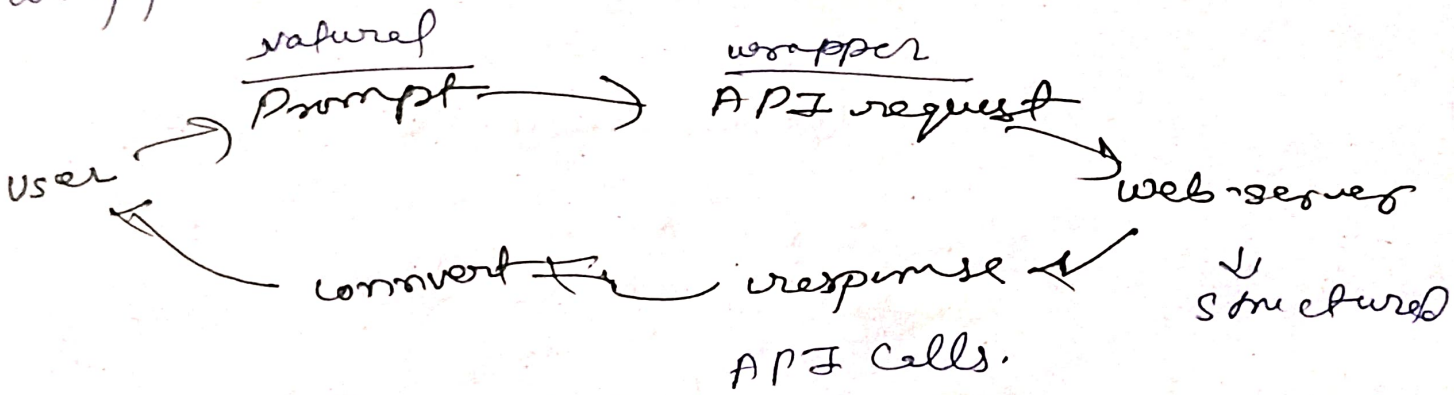
- Which base model to use
- What system instructions to inject.
- How responses should be structured.
- What constraints exists.
- What defaults should be applied.

And Ollama using the base model and instructions given creates a new model that have same brain but new ~~identity~~ identity

Ollama Using Rest API.

ollama → CLI commands, python library } wrappers around API

1) Ollama internally work using Rest API hosted locally at port 11434.
2) CLI commands and python library are just wrappers around the API end points.



LangChain

LangChain is an open-source framework that allows you to "chain" together different components to create advanced LLM application. It acts as an orchestration layer that manages how data flows between the user, the model (Ollama), and external tools (like Databases or Google Search).

- LLM →
  - → Memory
  - → Google Search
  - → Embedding
  - →

RAG

→ input handling
→ Chunking
→ Embedding
→ Database
→ ~~Relevant~~ Retrival
→ Generation

→ AI

Langchain [PyPDF]

[PyPDF → Chunks → ollama → FIAS → Retrival → Olla

Ollama Cloud

Sept - 2025 → ollama Cloud

[Ollama] → LLM → Hardware Constrains.

LLM → download → Local → Run
↳ working → Ram, VRAM.

Ollama Cloud is a cloud-based extension of the Ollama platform that lets you run large language models (LLMs) without needing powerful hardware. Instead of running on your own machine, the models are executed on powerful datacenter-grade hardware managed by Ollama.

[Large Parameter] → Ollama Cloud
↓
Ollama web